NeuroMPS

A provenance-tracked MEA data asset and a calibrated two-compartment digital twin

Team	NeuroMPS
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Research decision-support only. No efficacy, safety, diagnostic or clinical claim is made.

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Every number below carries an HTML comment naming the `outputs/metrics.json` key it came from. `python -m neuromps.report.verify` re-reads each citation and

fails if the document and the metrics disagree, so a figure that goes stale when a stage is regenerated cannot quietly survive in the prose. The decision record behind these results, including what was tried and abandoned, is `report/LAB_NOTEBOOK.md`.

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1. Summary

Two layers were built:

(A) A standardized, provenance-tracked data asset of public neural microelectrode-array (MEA) recordings: 21,272 records from four sources, each record carrying its source URL, license, SHA256 and download time, validated against a JSON Schema at a conformance level that requires every provenance hash to match the download manifest.

(B) A Brian2 spiking-network digital twin of an asymmetric two-compartment neural chip, calibrated against the asset, with simulation-based inference (`sbi`) attempting to recover connectivity and drug effects from MEA summary features.

The results that matter are as much negative as positive, and the report leads with both:

1. **Rate and burst-fraction measures survive a change of species; synchrony and network-burst measures do not** (§6.3, §6.5, §6.6, §7.3). **Across platforms none of them is safe** — mean firing rate itself reverses sign (§6.4), which is the sharper limitation of the two.
2. **Coupling asymmetry tracks relative firing rate, not pathway direction.** At fixed wiring, changing one chamber's drive moves the measure several times further than changing the wiring does (§7.2).
3. **The twin's posterior does not recover drug-knob direction from real recordings.** The inferred GABA-A scale shows no bicuculline dose trend, and untreated chips already infer 0.3781, so the parameter is absorbing model mismatch rather than tracking the compound (§7.5).
4. **Direction is recoverable from simulated features once firing rate is residualised out** — AUC 0.9359, 95% CI [0.893, 0.9739], from a 10 s recording on 24 electrodes per compartment — but as a contrast between two well-spaced wiring classes, not as a measurement of a continuous ratio (§7.6). A second arm, which adjusted drive to match rates, reaches a higher AUC at fewer electrodes and **fails its own validity check**; it is reported as inconclusive rather than as the result.

The honest summary of the twin is therefore: it reproduces baseline features and most drug directions, it can tell two wiring regimes apart in simulation, and it cannot yet read pharmacology off a real chip.

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2. Problem and scope

Microphysiological systems produce electrophysiology that is rich but hard to compare: different platforms, electrode counts, durations, species and feature defi-

nitions. Two consequences follow for anyone trying to use public MEA data for compound screening.

Data are not comparable by default. Feature names collide while definitions differ — “synchrony” from a spike-time tiling coefficient is not the Axion synchrony index — and provenance is usually lost by the time data reach an analysis.

A model of the chip is not a model of the measurement. A spiking network that reproduces a culture’s firing does not necessarily reproduce what 48 electrodes detect from it.

This entry addresses both: a schema and feature standard with enforced provenance, and a digital twin that is scored through the same feature extractor as the real recordings.

Out of scope. No clinical, diagnostic, efficacy or safety claim is made or implied. Compound labels are mechanism classes drawn from the source publications, not indications.

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3. The data standard

3.1 Record model and data levels

Records validate against `schema/neuromps_schema.json` (JSON Schema 2020-12). The central decision is that records carry an explicit **data level**, and that records only compare within a level:

level	meaning	records
`spike_times`	per-electrode spike trains; our feature vector is computable	897
`fitted_curve`	published concentration-response fits only	20,375
`per_well_features`	vendor feature tables without spike times	—

This distinction is load-bearing. Most of the asset by count is ToxCast curve fits, from which our nine-feature vector cannot be recomputed; only 897 records support the feature pipeline. A schema that hid that behind a single “features” field would invite exactly the wrong comparison.

3.2 Provenance and conformance

Conformance **level 2** requires that every record’s provenance hash matches `data/raw/MANIFEST.json`, and the build refuses to emit an asset otherwise. Each fetcher records URL, license, SHA256, byte count and download time, and re-checks the source’s advertised license at fetch time so a silently-relicensed upstream blocks rather than propagates.

3.3 Quality control

QC is recorded, never applied silently. Each record carries flags and a single `qc.usable` verdict, and the flags are informational: a flagged record is still in the asset, with the reason attached, so a downstream analysis can make its own decision.

flag	records	what it means
`incomplete_dose_series`	272	the compound's dose series is missing points
`too_few_active_electrodes`	169	under 10 electrodes cleared 5 spikes/min
`replicate_divergence`	151	one replicate departs from its group
`outlier_feature_value`	151	a feature sits far outside its source's distribution
`excluded_well_upstream`	90	the source's own log excluded the well
`noisy_electrodes_present`	11	the source marked electrodes as noisy

Two decisions here were driven by defects found in the data rather than by design. The replicate-divergence test originally used a robust z-score, which collapses when more than half a replicate group shares a value — common in sparse recordings, where the median absolute deviation is exactly zero; it now falls back to a mean absolute deviation and names the diverging feature. And the 90 wells excluded by an upstream log were at first missing from the asset altogether, because records were built from the feature tables alone; they are now built from the union of the log and the spike files, so an exclusion is represented rather than absent.

3.4 Feature set

Nine features form the core vector, computed by a single extractor (`features/mea.py`) that the twin must also call:

feature	definition as computed
`mfr_hz`	spikes per second per active electrode
`burst_rate_per_min`	bursts per minute per active electrode
`burst_duration_s`	mean burst duration
`spikes_per_burst`	mean spikes in a burst
`pct_spikes_in_bursts`	share of spikes falling inside bursts
`network_burst_rate_per_min`	network bursts per minute, chance-corrected participation
`synchrony_index`	spike-time tiling coefficient, mean over electrode pairs
`isi_cv`	coefficient of variation of inter-spike intervals
`active_electrodes`	electrodes clearing the activity threshold

Thresholds are fixed and recorded per record: an electrode is active at ≥5 spikes/min, a well is usable at ≥10 active electrodes, a burst is ≥5 spikes with ISI ≤0.1 s, and the tiling coefficient uses $\Delta t = 0.05$ s.

The network-burst threshold deserves its own note, because the obvious implementation is wrong. A fixed participation threshold — n electrodes firing in the

same window — counts coincidences that independent electrodes produce anyway; at 5 Hz on 16 electrodes that is on the order of fifty spurious "network bursts" per minute, which would dominate the measure. The threshold is therefore set per recording so that the expected number of chance detections stays below 0.05, from the recording's own electrode count and firing rates. This matters beyond bookkeeping: because the correction scales with electrode count, it is a candidate explanation for part of the cross-platform disagreement reported in §6.4, and that remains untested.

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4. The data asset

source	records	level	species	license
`toxcast_mea`	20,375	fitted_curve	rat	CC0-1.0
`screenneuropharm_bic`	580	spike_times	rat	MIT
`kapucu2022`	288	spike_times	human, rat	CC-BY-4.0
`sharf2022`	29	spike_times	human	CC0-1.0

Totals: 21,272 records, of which 20,831 are marked usable and 441 are not. By species, 21,099 records are rat and 173 are human.

The species imbalance is the asset's main structural weakness and is stated wherever human results appear: every human-species conclusion rests on 173 records from two sources.

4.1 A rejection, as a worked example of the license discipline

Tukker 2020 (rat and hiPSC MEA pharmacology) is a good fit scientifically and is **excluded**. It is CC BY-NC 4.0, and the competition rules assign joint ownership of submissions rather than preserving an entrant's own license, so the non-commercial condition cannot be guaranteed downstream. It is recorded as `skipped` in the manifest with that reason rather than dropped silently. The cost is real: without it, the rat→human transfer study rests on one source.

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5. Methods

5.1 Cross-validation and resampling

Every model is evaluated with **grouped** cross-validation, grouping by culture/well so replicates of one culture never span the train/test boundary, and every interval is a **cluster bootstrap over those groups**, never over rows — with 1,000 resamples in a full run and 200 in `--quick`, the count recorded alongside each estimate. Bootstrapping rows here would report intervals roughly a factor of two too narrow, because rows within a culture are not independent.

5.2 Tasks

- **Task 1 — mechanism classification** (within platform): predict mechanism class from the nine-feature delta vector of a treated/baseline pair.
- **Task 2 — external detection**: apply the Task 1 model to a source it never saw, with a negative control of the opposite mechanism.
- **Task 3 — platform contrast**: compare the same mechanism measured on two platforms.
- **Transfer**: rat → human, with and without domain adaptation.

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6. Results: data and models

6.1 Task 1 — mechanism classification

Six classes, 73 treated/baseline pairs across 50 culture groups, 4-fold grouped CV.

model	balanced accuracy	95% CI
logistic regression	0.537	[0.4216, 0.6845]
gradient boosting	0.398	[0.2762, 0.5178]
most-frequent baseline	0.167	—

Chance is 0.167. Logistic regression beats chance with an interval that excludes it. Its interval and gradient boosting's overlap heavily, so the gap between the two models is within noise; with 73 pairs the simpler model is the one to prefer, not the one shown to be better.

Calibration. Expected calibration error on out-of-fold predictions is 0.1114, 95% CI [0.0773, 0.2557] (`outputs/figures/task1_reliability_real.png`). The upper end of that interval is large: predicted probabilities should be read as ordering, not as calibrated risk — and not even as ordering above 0.8 confidence, where observed accuracy falls rather than rises, on a handful of predictions.

Conformal sets. At a 95% target the empirical coverage is 0.918 with a mean set size of 4.562 classes out of six — honest coverage at the cost of sets too large to act on. That is the correct reading of 73 calibration points, not a defect to tune away.

6.2 Task 2 — external detection and its negative control

Applied to bicuculline recordings from a source the model never saw, the GABA-A antagonist class is detected in 0.366 of 497 records, and the detection does **not** track dose (Spearman 0.014, p = 0.751).

The negative control behaves as it should: diazepam, a positive modulator rather than an antagonist, is called GABA-A antagonist in only 0.077 of its records.

This is a weak result reported as a weak result. A detection rate near a third with no dose relationship is not a screening tool; it is evidence that the mechanism signal partially survives a platform change.

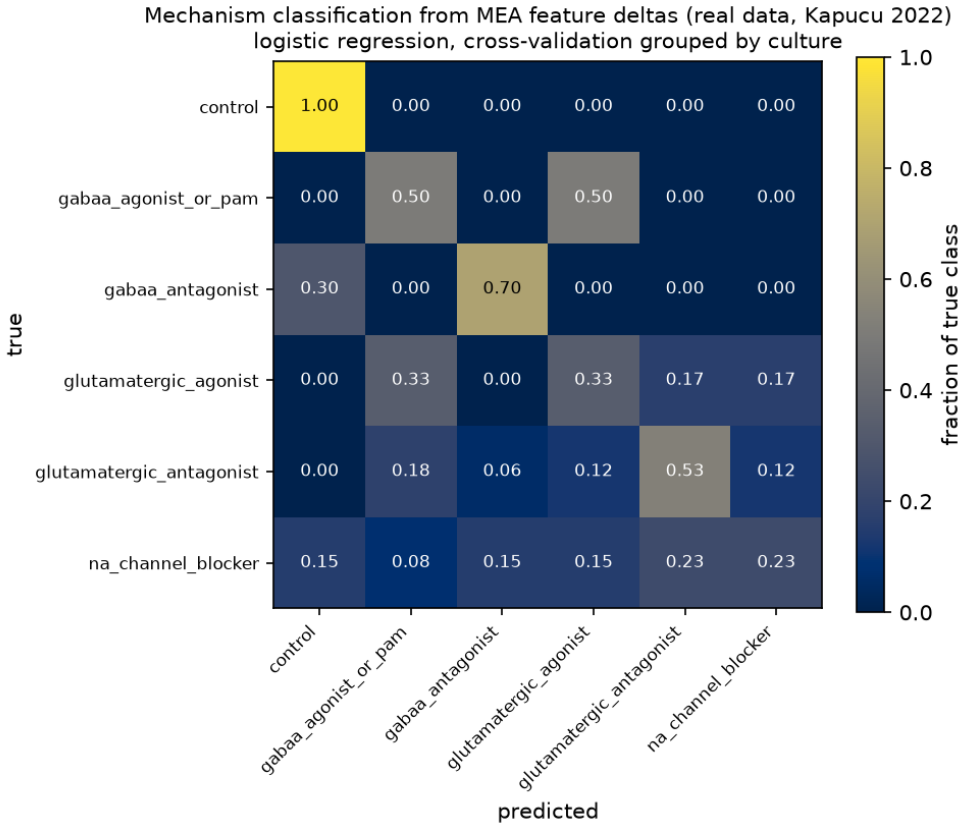


Figure 1: Task 1 confusion matrix, REAL data: out-of-fold predictions from grouped cross-validation over 73 treated/baseline pairs. Errors concentrate between the two GABA-A classes.

6.3 Species transfer (rat → human)

setting	balanced accuracy	95% CI
human, within-species	0.646	—
rat, within-species	0.543	—
rat → human, no adaptation	0.55	[0.4, 0.6]
rat → human, CORAL	0.442	[0.28, 0.597]
rat → human, per-species standardisation	0.408	[0.246, 0.56]

Chance is 0.2. Transfer works above chance (permutation p = 0.0002), and **both adaptation methods made it worse**. With 37 training and 32 test records, CORAL is estimating a covariance it cannot support. Reporting adaptation as a negative result is the useful outcome here.

6.4 Task 3 — platform contrast

The same mechanism at ≥30 µM measured on two platforms: gabazine on 16-electrode Axion wells, n = 10, against bicuculline on multi-electrode arrays, n = 167. Five features agree in sign and four disagree.

The disagreement does not fall where the species analysis puts it. `mfr_hz` reverses, +0.348 against −0.684, while `synchrony_index` agrees, +1.743 against +0.148. That is the opposite of the pattern in §6.5 and §6.6, and it is the reason this report separates the two claims: **the portable-feature result is about species, and does not carry across platforms.**

Three cautions on how far this goes. The contrast confounds compound (gabazine against bicuculline) with electrode count, recording duration and culture age, so it bounds portability rather than explaining it. The two arms are ten wells against

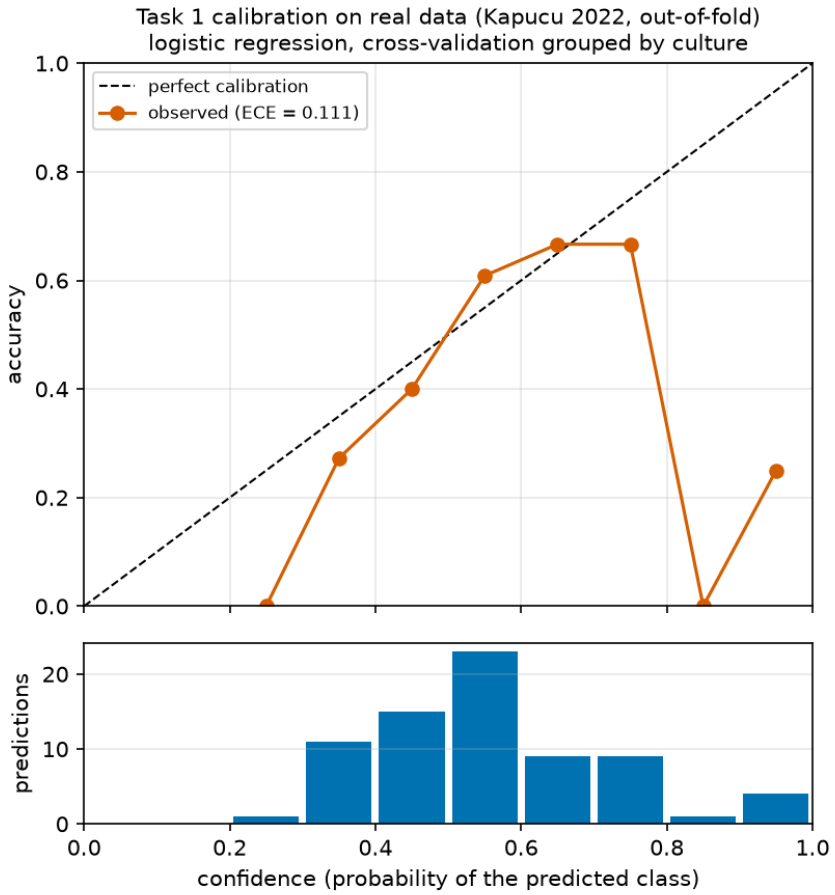


Figure 2: Reliability diagram, REAL data: predicted probability against observed frequency for the mechanism classifier, out-of-fold. Bin counts are small, which is why the calibration interval is wide.

feature subset	balanced accuracy	95% CI
all nine	0.514	[0.3983, 0.6619]
rate only	0.53	[0.3964, 0.6235]
species-invariant three	0.516	[0.371, 0.6295]
synchrony only	0.455	[0.3554, 0.586]
network measures only	0.417	[0.2983, 0.5573]

Two rate features match all nine. Leave-one-out agrees: dropping `mfr_hz` changes balanced accuracy by −0.05, while dropping `burst_rate_per_min` changes it by 0.0.

6.7 Compartment asymmetry in real devices

Across 16 two-chamber devices, the statistic that separates device types is coupling **lag**, not asymmetry. Median absolute lag is 11.25 ms for cortical–hippocampal devices against 5.0 ms and 5.0 ms for the two same-tissue types (Kruskal p = 0.0082; Holm-adjusted 0.0413 and 0.0257 against the two homogeneous types).

The internal control behaves. The two same-tissue types are indistinguishable from each other, Holm 0.4785, which is what a real device-type effect should look

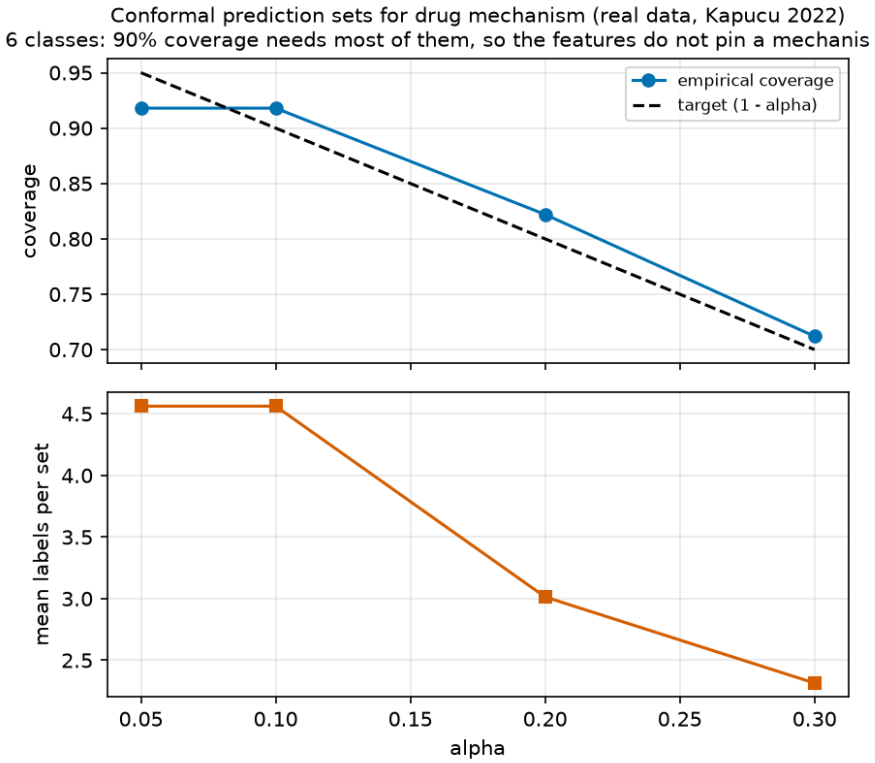


Figure 3: Conformal set size against target coverage, REAL data. Honest coverage costs sets of four to five classes out of six.

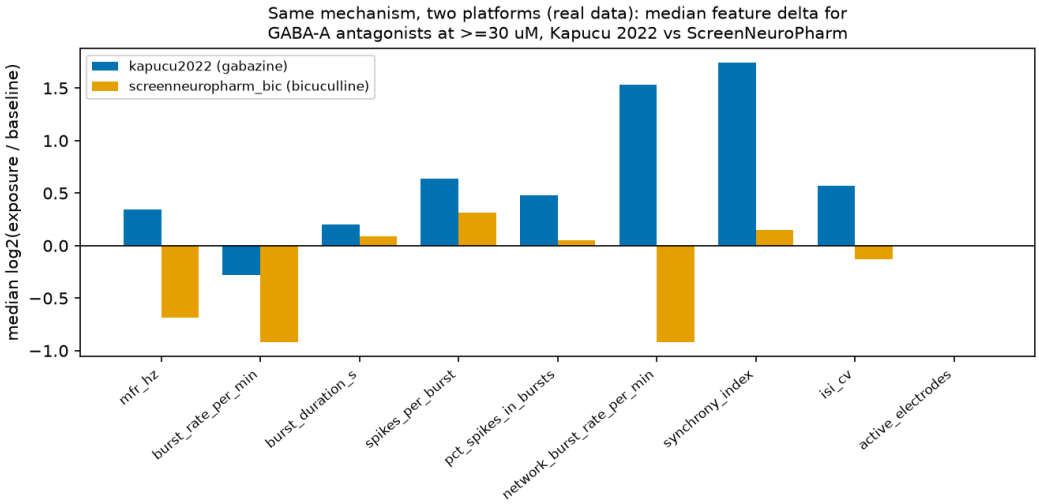


Figure 4: The same mechanism measured on two platforms, REAL data: per-feature log2 change for GABA-A antagonists. Mean firing rate reverses sign across platforms; synchrony agrees.

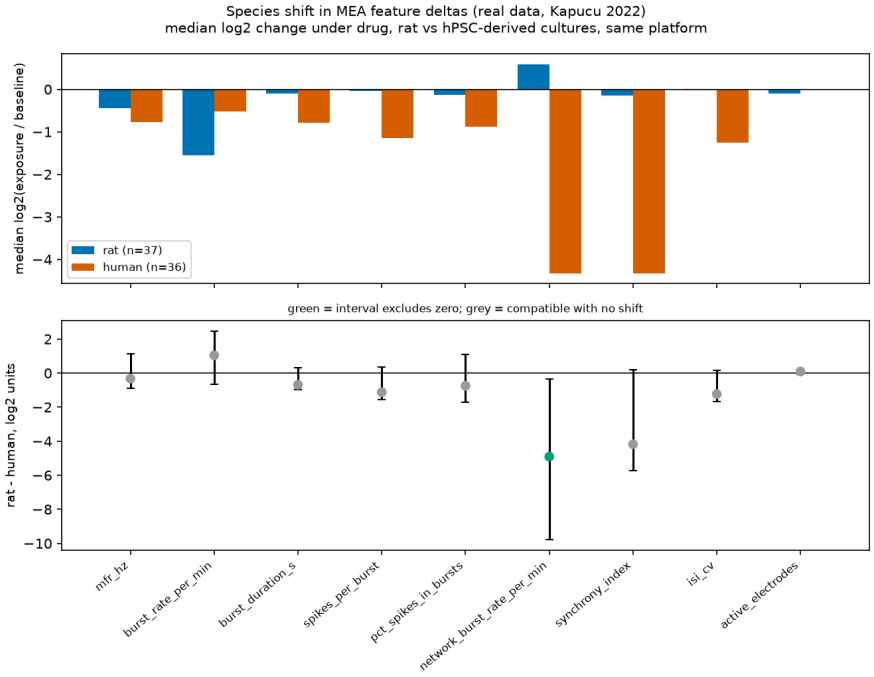


Figure 5: Per-feature species shift, REAL data: rat against human baseline distributions, with cluster-bootstrap intervals. Rate and burst-fraction features move least; synchrony moves most.

like and what a pooled heterogeneous-against- homogeneous split cannot show.

Two things this section does **not** claim. On absolute asymmetry rather than lag, the same per-type comparison does not survive correction, Holm 0.0519, so lag is the claim and asymmetry is not. And a pooled heterogeneous-against-homogeneous test of asymmetry does cross the line, p = 0.042; it is kept in `metrics.json` and is deliberately not the headline here, because pooling CxCx with HpHp is an artefact — the two homogeneous types differ from each other on peak correlation, so pooling them manufactures a contrast. An earlier draft of this report led with that pooled test; the lab notebook had already recorded it as abandoned, and this section now reports the analysis that replaced it.

With five to six devices per group, this supports an ordering of device types, not an effect size, and §7.2 gives a specific reason to distrust the asymmetry measure in particular.

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7. Results: the digital twin

7.1 Model

Two compartments of conductance-based leaky integrate-and-fire neurons with spike-frequency adaptation, 80/20 excitatory/inhibitory, recurrent intra-compartment connectivity and **directional** inter-compartment projections whose forward/backward ratio is the parameter of interest. Poisson background drive. Connectivity is parameterised by **in-degree rather than probability**, so a calibration done at one network size still holds at another.

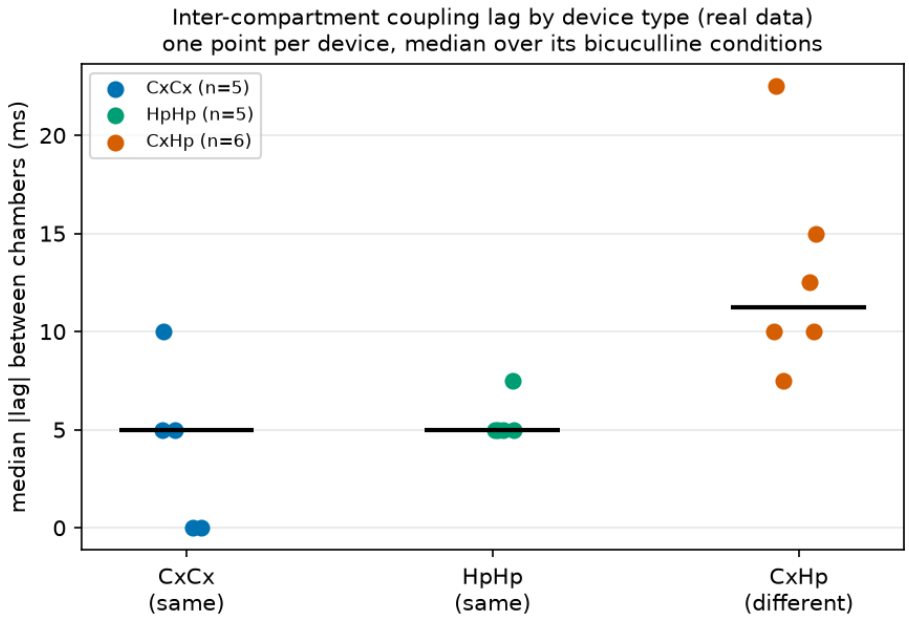


Figure 6: Inter-compartment coupling lag by device type, REAL data, one value per device: five CxCx, five HpHp, six CxHp. The two same-tissue types are indistinguishable from each other; the heterogeneous type sits apart.

The parameters carried by the network, and how each was fixed:

parameter	role	how it was set
`intra_in_degree`	recurrent inputs per neuron	calibrated, searched over 10-35
`inter_in_degree`	cross-compartment inputs per neuron	inferred, prior 4-16
`forward_ratio`	forward against backward pathway strength	inferred, prior 1-16 in log space
`excitatory_weight`, `inhibitory_weight`	synaptic conductance steps	calibrated
`background_rate_hz`, `background_weight`	Poisson drive	calibrated; drive to the target chamber is a separate scale
`adaptation_increment`, `adaptation_tau_ms`	spike-frequency adaptation	calibrated
`threshold_spread`	per-neuron excitability heterogeneity	calibrated, searched over 0-0.3
`detection_radius_um`	how far an electrode hears	calibrated, searched over 100-300 µm
`gabaa_scale`, `ampa_scale`, `threshold_scale`	drug knobs, 1.0 is untreated	inferred

Identical neurons synchronise almost perfectly, which no real culture does, so `threshold_spread` exists to stop the twin being trivially more synchronous than

any recording. `detection_radius_um` is a property of the virtual electrode rather than of the culture, and is called out as such in the calibration record so it is not read as a biological parameter.

Neurons are placed on a virtual electrode grid and only spikes within a detection radius reach an electrode; the resulting per-electrode spike trains go through the same `features/mea.py` as real recordings, and — after a defect found in HD16 — through the same coupling measure on the same per-electrode input.

7.2 The measure the twin exists to use is confounded

Coupling asymmetry was expected to report pathway direction. It reports relative firing rate. At **fixed** wiring, lowering the receiving chamber's background drive moves asymmetry from 0.109 to 0.681; at **fixed** drive, raising the forward ratio from 1 to 16 moves it only from 0.109 to 0.143. The measure's own range is [−1, 1], so this is not the whole of it — but drive moves the statistic several times further than the wiring it is supposed to be reporting. This is the most consequential single finding about the twin, and it propagates into every later section: any analysis that reads direction off coupling must control for rate first.

Three checks on the real devices confirmed the confound does not obviously bite there (firing-rate imbalance does not differ by device type), which is why §6.6 survives as a hypothesis.

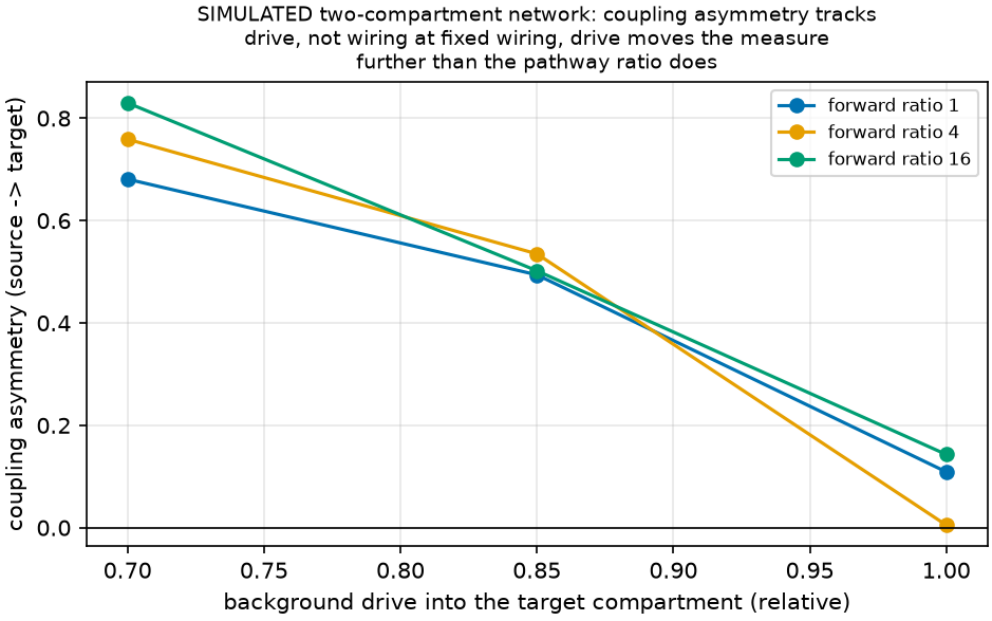


Figure 7: SIMULATED: measured coupling asymmetry against the forward/backward wiring ratio, at three levels of drive to the receiving chamber. Drive moves the measure further than wiring does.

7.3 Calibration

A random search over seven tuned parameters brought the simulated baseline into the real band for `mfr_hz`, `burst_rate_per_min` and `network_burst_rate_per_min`, and **not** for `pct_spikes_in_bursts` or `synchrony_index`. The residual mismatch is precisely the burst-structure and synchrony group that §6.5 and §6.6 independently

identify as the unreliable part of the feature set across species. Three analyses agree about species; the platform contrast of §6.4 disagrees, and that disagreement is itself the finding.

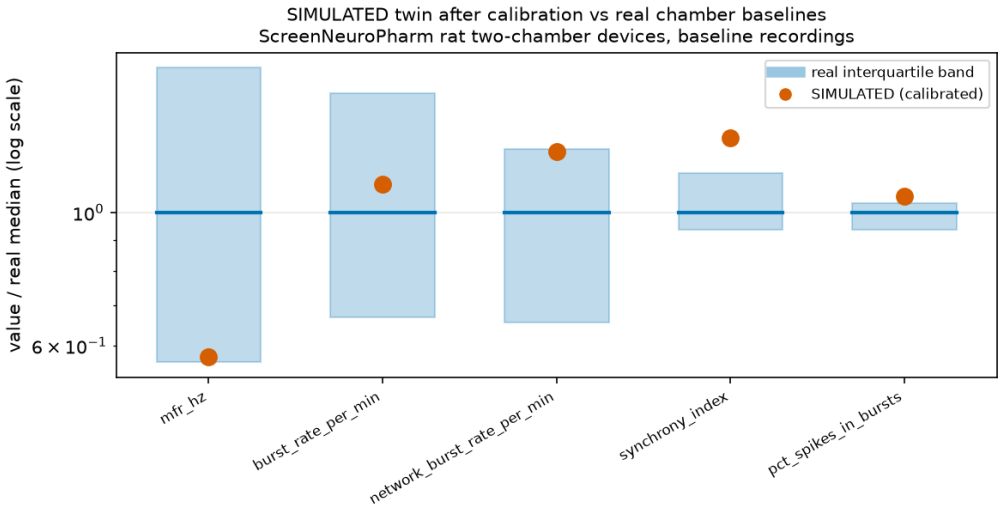


Figure 8: SIMULATED against REAL: calibrated baseline features of the twin compared with the real baseline band. Rate and burst-rate features land inside it; burst fraction and synchrony do not.

7.4 Drug knobs

Each knob is compared against the real direction for its mechanism, scoring only features whose real direction is established by a cluster bootstrap:

knob	agrees / scored
GABA-A block	4 / 5
GABA-A potentiation	2 / 4
AMPA block	4 / 4
sodium-channel block	3 / 3

Directions only: each knob is compared against a real reference from a different platform, so this supports direction, never magnitude.

7.5 Simulation-based inference, and what it cannot do

A neural posterior estimator was trained on 3,500 simulations.

parameter	posterior contraction	KS statistic
`threshold_scale`	0.8	0.0571
`ampa_scale`	0.443	0.0871
`gabaa_scale`	0.323	0.1093
`inter_in_degree`	0.32	0.1029
`log10_forward_ratio`	0.311	0.0957

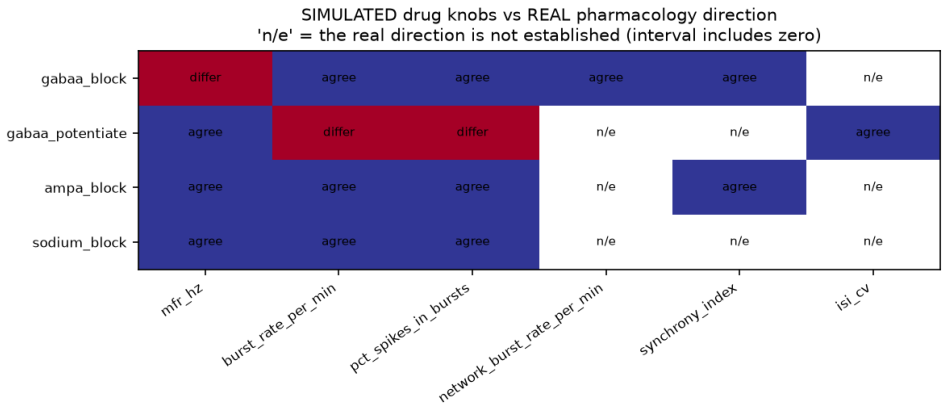


Figure 9: SIMULATED drug knobs against REAL pharmacology directions, per feature. Only features whose real direction is established by a cluster bootstrap are scored.

Only excitability is well identified, and only `threshold_scale` passes a rank-uniformity check ($p = 0.1956$; every other parameter is below 0.01). **Posterior intervals from this estimator should not be quoted as calibrated.**

That statement is weaker than it looks, and the reason is worth recording. Posterior training was unseeded until this report was being built: rerunning identical code on the identical simulation cache moved contraction by about 0.02 and moved this p-value between 0.46, 0.0092 and 0.1956 — on one of those runs `threshold_scale` failed the check that it passes here. Training is now seeded and verified to reproduce across processes, but the run-to-run spread is the honest measure of how much weight a single rank-uniformity p-value can carry. Contraction was stable across the same runs; the rank test was not. A learning curve showed connectivity contraction still rising with training-set size but only slightly: more simulations alone will not fix it.

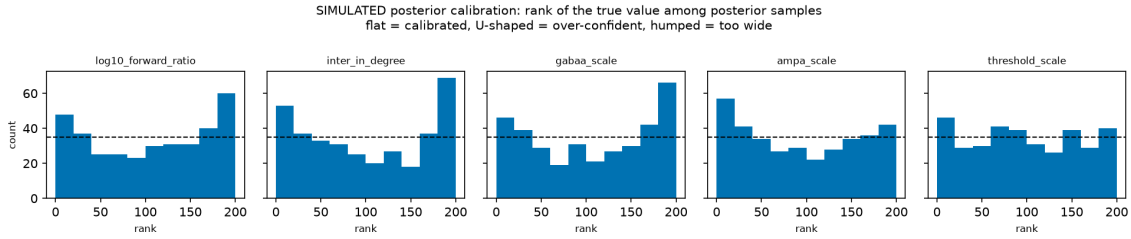


Figure 10: SIMULATED posterior calibration: rank of the true parameter among posterior samples. Flat is calibrated, U-shaped is over-confident. Only `threshold_scale` is close to flat.

Applied to 113 real two-chamber recordings from 16 devices, with 99.2% of real summary values inside the simulated range:

- The inferred GABA-A scale shows **no dose trend**: Spearman 0.118, 95% CI [−0.168, 0.419], clustered over devices.
- **Untreated recordings already infer 0.3781**, CI [0.3149, 0.4677], excluding the neutral value of 1.0. The knob is absorbing general model mismatch.
- Four summary dimensions — burst rate and active electrodes in both chambers — pass a range check but have almost no simulated draws where the real values sit. Range containment is not density containment.

An earlier version of this analysis produced an apparent dose trend ($\rho = -0.197$, $p = 0.036$). It was an artefact of pseudo-replication — 113 recordings from 16 devices treated as independent — and of posterior extrapolation on out-of-distribution dimensions. Both are now checks in code.

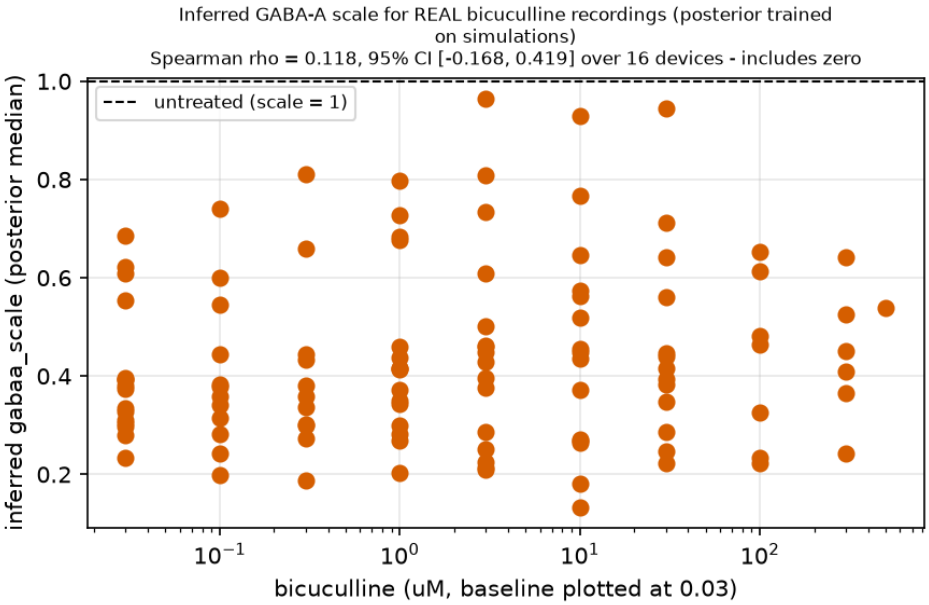


Figure 11: Inferred GABA-A scale for REAL bicuculline recordings, from a posterior trained on SIMULATIONS. There is no dose trend, and untreated chips already sit far below the neutral value of 1.

7.6 Directionality: what it takes to see asymmetry

120 simulated networks per arm with known wiring, symmetric against asymmetric. Electrode count is varied by subsampling one fixed array and recording length by truncating one simulation, so neither varies the culture.

The naive version is confounded exactly as §7.2 predicts: the full feature set separates the classes at AUC from 0.9219 to 0.9863, and three firing-rate features alone reach from 0.8644 to 0.9841.

Residualising every feature against both chambers' firing rates removes that, and carries its own validity check: rate features after residualising score from 0.3869 to 0.5969 and do not clear chance in every cell. Direction still separates at 0.9359, 95% CI [0.893, 0.9739], from a 10 s recording on 24 electrodes per compartment.

The rate-matched arm does not pass its own check, and is not the result. `metrics.json` states the criterion: in that arm, rate-only features should fall towards chance, and if they do not, the matching failed and the arm says nothing about direction. They do not — rate-only AUC there runs from 0.8587 to 0.9859. The drive adjustment did not remove the rate imbalance so much as reverse its sign, from −0.2756 and −0.5180 before to +0.3116 and +0.2377 after, leaving a worst relative gap of 0.26 against 0.2248 before it. That arm reaches AUC 0.9784 on only 6 electrodes, and an earlier draft of this report quoted exactly that number as the headline. It is withdrawn: the guardrail written into `metrics.json` says the arm is uninterpretable, and the correct response is to report the arm that passes, at the electrode count it actually needs.

Read this as a contrast, not a measurement. The same summaries that classify two well-spaced wiring regimes at 0.98 give a posterior contraction of only 0.311 for the continuous ratio (§7.5). Classifying is a far weaker demand than estimating.

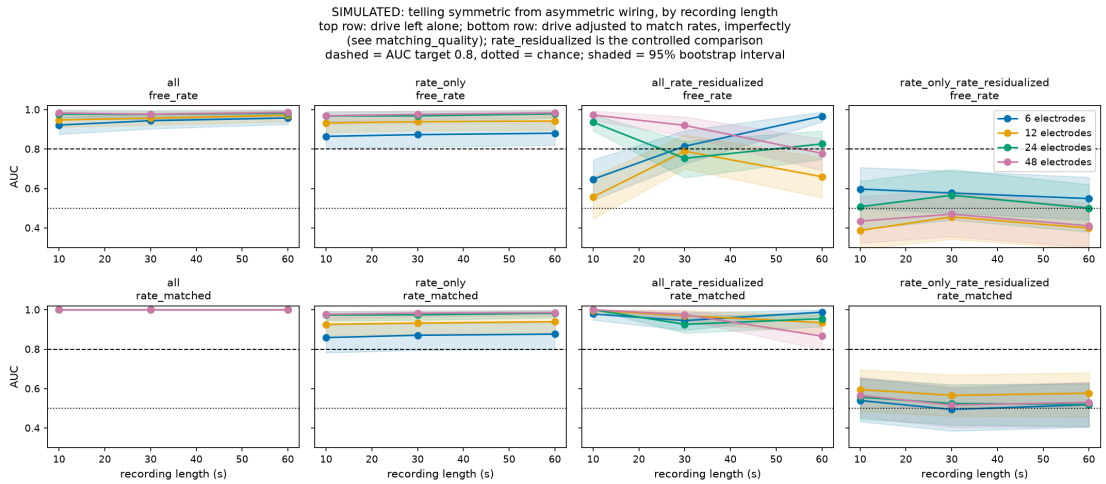


Figure 12: SIMULATED: telling symmetric from asymmetric wiring, by recording length and electrode count. Top row leaves drive alone; the bottom row adjusts drive to match rates and does so imperfectly, which is why that arm is reported as inconclusive. The rate-residualised columns are the controlled comparison, and the rightmost column is the validity check itself.

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8. Limitations

Data. The asset is 21,099 rat records against 173 human; every human claim rests on two sources. Most records are curve fits from which our features cannot be recomputed. Dropping Tukker 2020 on license grounds leaves the transfer study on a single source. Compound coverage is thin: six mechanism classes, 73 pairs.

Models. Balanced accuracies around 0.5 on six classes are above chance but far from usable; intervals are wide and overlapping. Conformal sets cover honestly but are too large to act on. Calibration error may be as high as 0.2557. The nine-feature vector is effectively a rate feature plus noise for this task. The implementation uses single-level grouped CV where the plan specified nested CV, so hyperparameter choice is not fully held out.

The twin. Coupling asymmetry tracks rate, not direction (§7.2). Calibration misses burst structure and synchrony (§7.3). Connectivity parameters are weakly identified and no parameter's posterior is calibrated except by the weakest test (§7.5). On real data the drug knob absorbs model mismatch (§7.5). Simulated recordings are 10 s against real recordings of 480 s. The simulated array is one fixed geometry while real chambers carry 48, 36 or 60 electrodes. HD16's positive result is identifiability under a correct model, since the same simulator generated both classes, and its rate controls remove only linear rate dependence.

Process. A CI workflow is configured and the work described here has been pushed to it, but no CI result has been observed from the environment these results were produced in; treat CI as unverified. The demo video specified for Phase 4 has

not been recorded. Every twin metric produced before the coupling-definition fix is incomparable with those after it, and was regenerated.

A fuller list, with dates and commits, is in `report/LAB_NOTEBOOK.md` (33 numbered limitations).

---

9. Reproducibility

```
pip install uv
uv venv --python 3.11 .venv
uv pip install --python .venv/bin/python -e ".[dev]"
python run_all.py --quick     # every figure and metric, reduced settings
python run_all.py             # full run
```

- `--quick` completes in about 470 s on CPU against a 600 s budget, and each run records its own per-stage timings under `metrics.run_all_timings_quick`; the full twin stage takes roughly 2,000 s and records `metrics.run_all_timings_full`. Both are logged under `outputs/logs/` as well.
- **`--quick` reproduces every figure and metric except the §7.6 directionality sweep.** At reduced settings that sweep has too few simulations per cell to fit a discriminator, so every `cheapest_configuration` entry is written as `null` and the validity check is not meaningful; the §7.6 numbers need `python run_all.py --stage twin` at full settings. `metrics.json` records which mode produced the stored value (`twin_hd16_directionality.quick`), and the quick run also evaluates 60 of the 350 posterior holdout points rather than all of them, recording the count it used.
- Dependencies are pinned in `pyproject.toml`; numpy is held at 2.3.5 because brian2 2.9.0 does not import against 2.4.
- Every number in this report is produced by a script under `src/` and written to `outputs/metrics.json`; `python -m neuromps.report.verify` checks the report against it.
- Caches refuse to be reused across a change in what they mean: the simulation cache stores its summary names, electrode count and a hash of the source of every function that defines a summary.
- Tests run with `pytest` (feature extraction, QC flags, provenance hashing, the MEA simulator, and a contract test that fails if any headline metric loses its confidence interval).

9.1 Where the time goes

Each run writes its own stage timings, so this table is generated rather than transcribed: see `metrics.run_all_timings_quick.by_stage_s` for the quick run and `run_all_timings_full` for a full one. In the runs behind this report the quick pipeline spends most of its time in the twin stage (validation, calibration, drug knobs, inference, real-data inference, directionality) and in the demo precomputation, with fetch varying by about a minute between runs depending on how quickly the license endpoints respond — which is the main reason the budget headroom is not stable.

9.2 What the repository enforces, rather than documents

- **Provenance.** The asset build fails if any record's provenance hash disagrees with the manifest.
- **One feature extractor.** The twin computes features through the same `features/mea.py` as real recordings, and — after a defect that survived for several phases — through the same coupling measure on the same per-electrode input.
- **Interval discipline.** A contract test fails if any headline metric loses its confidence interval, and the resample count is recorded next to every estimate.
- **Cache meaning.** The simulation cache stores its summary names, its electrode count and a hash of the source of every function that defines a summary; it refuses to load across a change in any of them.
- **Report numbers.** `neuromps.report.verify` re-reads every citation in this document and fails on disagreement.
- **No clinical claims.** A test scans the reports for efficacy, diagnostic and treatment language and fails unless the line is a disclaimer ruling it out.

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10. The demo

`streamlit run app/app.py`, after `python run_all.py --quick`, opens three tabs — the data asset and its standard, mechanism prediction with a conformal set beside the transfer results, and the twin with its posterior and what it does on real recordings.

The design constraint was that an interaction must never wait on a simulator. `run_all.py --stage demo` precomputes a grid of 75 twin simulations spanning the prior, with posterior samples per point, posterior samples for all 113 real two-chamber recordings, and a classifier fitted once, in 85.4 s. Measured through Streamlit's own test harness, every interaction lands between 0.6 and 0.9 s; the one exception is a button that runs a single Brian2 simulation at the current slider values in about 4.5 s, which exists so that the twin is verifiably a simulator rather than a lookup table. Timings and the tab-by-tab description are in `app/README.md`; screenshots of all three tabs are in `outputs/figures/app_tab*.png`.

The app reads `outputs/metrics.json` and `outputs/demo/` and recomputes nothing, so it cannot disagree with this report.

11. AI assistance disclosure

This work was produced in a paired workflow with Anthropic's Claude (Claude Opus 5) via Claude Code. The model wrote the large majority of the code, tests and prose in this repository, under step-by-step human direction: the human set the plan and phase gates, chose the scope, approved or rejected each step, and directed the scientific decisions.

Specific practices used to keep that honest, all visible in the repository history:

- Numbers are never written by hand into documents. Every figure in this report is emitted by a script into `outputs/metrics.json` and re-checked against

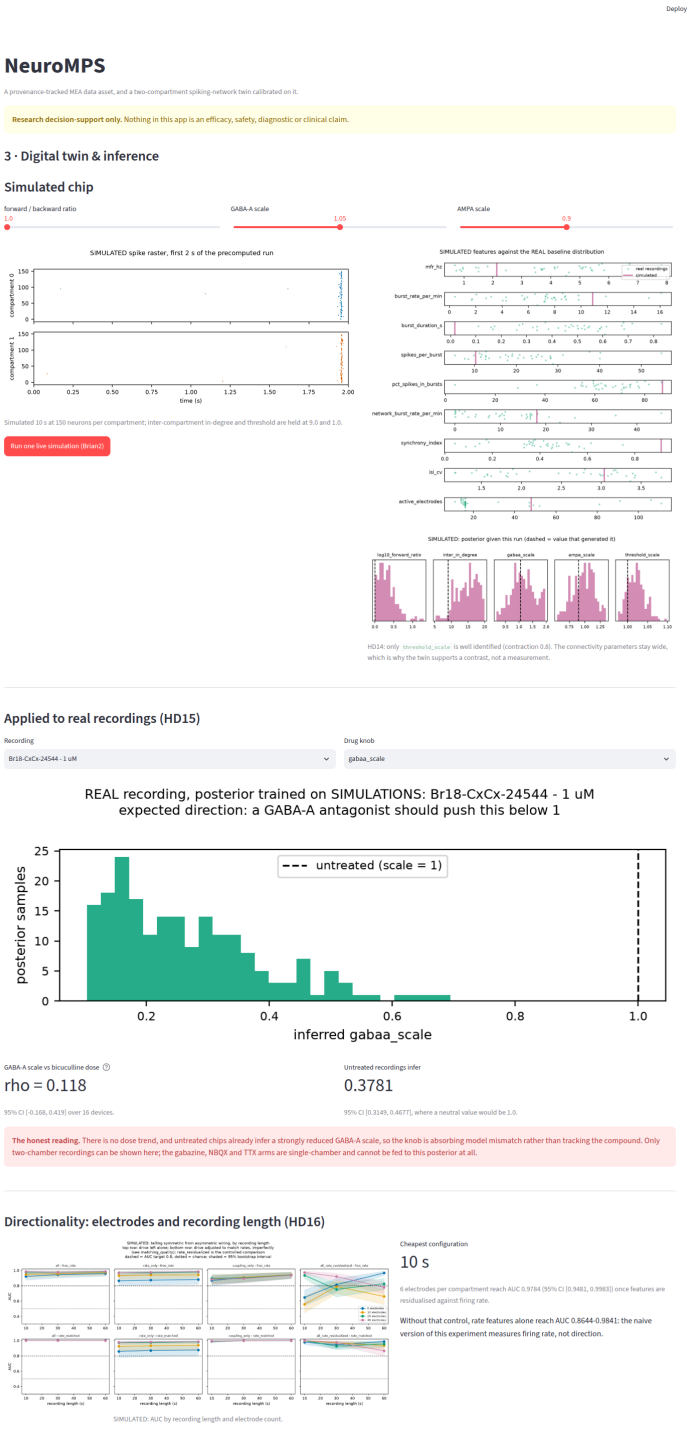


Figure 13: The demo's third tab: a SIMULATED twin grid with its posterior marginals, above the HD15 result on REAL recordings and the SIMULATED HD16 directionality panel.

it by `neuromps.report.verify`, which caught a stale timestamp and three misanchored citations while this report was being written.
- Results were reported as observed, including when they contradicted the plan's expectation — the negative findings in §6.2, §6.3, §7.2 and §7.5 are examples.
- `report/LAB_NOTEBOOK.md` records what was tried and abandoned, with dates and commit hashes, including several defects the model introduced and then found in its own work.

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12. Sources and licenses

Data sources, with licenses and citations, are listed in `CITATIONS.md` and recorded per file in `data/raw/MANIFEST.json`. Third-party software licenses are in `THIRD_PARTY.md`. This repository is released under the MIT license (`LICENSE`).

Appendix A. Where each result lives

Every number in this report comes from a key in `outputs/metrics.json`, written by a script under `src/`. The main entry points:

result	metrics key	script
asset counts and provenance	`data_asset`, `data_card`	`neuromps.etl.build`
mechanism classification, calibration	`mechanism_baselines`	`neuromps.models.mechanism`
conformal sets, ablations	`uncertainty_and_ablations`	`neuromps.models.conformal`, `.ablation`
species transfer and shift	`species_transfer`	`neuromps.models.transfer`
compartment asymmetry	`compartment_asymmetry`	`neuromps.models.asymmetry`
twin baseline and drive sweep	`twin_hd11_validation`	`neuromps.twin.validate`
twin calibration	`twin_hd12_calibration`	`neuromps.twin.calibrate`
drug knob directions	`twin_hd13_perturbation`	`neuromps.twin.perturbation`
posterior training and coverage	`twin_hd14_inference`	`neuromps.twin.inference`
posterior applied to real recordings	`twin_hd15_real_inference`	`neuromps.twin.real_inference`
directionality experiment	`twin_hd16_directionality`	`neuromps.twin.directionality`

Figures are written to `outputs/figures/` (real data) and `outputs/twin/` (twin output). A twin figure carries `simulated` in its filename unless it plots **real** recordings under a simulation-trained model — `hd15_real_dose_response.png` is the one such hybrid, and its caption names both sides. The rule that matters is that no simulated output is ever presented as measurement, and that is enforced in the captions, not the filenames.

Appendix B. Reading the negative results

Three results in this report are negative, and each is reported because the alternative would have been to quote a number that does not mean what it appears to mean.

An apparent dose trend that was not one. The first pass at §7.5 gave Spearman $\rho = -0.197$ with $p = 0.036$. Two things were wrong: 113 recordings come from 16 devices, so a p-value across recordings treats replicates as independent; and the posterior was extrapolating on summary dimensions the simulator cannot produce. After a cluster bootstrap over devices and a sensitivity check that neutralises those dimensions, nothing remained.

A discrimination that was reading firing rate. In §7.6 the full feature set separates wiring classes at AUC 0.98 — and so do three firing-rate features alone. Two attempts to fix this by matching the drive both failed, each trading one rate mismatch for another. Residualising features against both chambers' rates is what worked, and it carries a validity check: rate features, after residualising, score no better than chance.

A calibration that misses exactly the features that do not transfer. The twin's residual mismatch in §7.3 is burst fraction and synchrony — the same group that shifts most between species in §6.5 and contributes nothing in the ablations of §6.6. Three analyses, arrived at separately, agree about species. The platform contrast in §6.4 points the other way, and that is why the portability claim in this report is restricted to species.